

Software Analyst Document

Project: Aim Fitness Center – Offline Gym Management System

1. Requirement Analysis

The Software Analyst studies the Requirement Gathering Document and converts business requirements into software modules. Since the application is fully offline, all data is stored locally and processed without an internet connection.

2. Business Flow

Login → Dashboard → Member Registration → Membership Plan → Attendance → Workout Plan → Progress Tracking → Membership Renewal → Reports

3. Screen Planning

Login, Dashboard, Member Registration, Member List, Membership Management, Trainer Management, Attendance, Workout Plan, BMI Calculator, Reports, Backup & Restore.

4. Data Collected

Member ID, Name, Age, Gender, DOB, Blood Group, Height, Weight, Phone, Email, Address, Emergency Contact, Membership Plan, Join Date, Expiry Date, Assigned Trainer.

5. Data Displayed

Member Profile, Active/Expired Membership, Attendance History, Trainer Details, Workout Schedule, BMI, Progress History, Dashboard Statistics.

6. UI Components

Text Boxes, Date Picker, Combo Box, Radio Buttons, Check Boxes, Tables, Search Box, Action Buttons, Menu Bar, Dashboard Cards.

7. Stakeholder-Based Design

Receptionist – Desktop Application

Trainer – Desktop Application

Gym Manager – Desktop Dashboard

Owner – Reports Dashboard

8. Modules

Authentication, Member Management, Membership Management, Trainer Management, Attendance Management, Workout Plan, BMI Calculator, Reports, Backup & Restore.

9. System Design

Java Swing desktop interface with a local database (SQLite/MySQL Localhost). All reports are generated locally and data remains within the gym computer.